

Neutrinoless Double-Beta Decay in BST: A Sharp Prediction

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⌚ REVISED 2026-07-18 — PREDICTION REVERSED (Dirac $\rightarrow$ Majorana). This paper originally predicted *Dirac* neutrinos and $|m_{\beta\beta}| = 0$ ($0\nu\beta\beta$ forbidden). On 2026-07-16 BST reversed this to Majorana (F413/K673, Toy 4691; Weinberg operator + odd-g chirality lock forbidding a unitary light ν_R). The current BST prediction is that $0\nu\beta\beta$ OCCURS, $|m_{\beta\beta}| \in [1.44, 3.63]$ meV (normal ordering, $m_1 = 0$), and a detection supports BST; a confirmed null below ~ 1 meV would falsify. The Abstract and Sections 2, 4, 6, 9 below are revised to this reading; the old Dirac argument is preserved as the retired reading (§4.2) with the topological reframe (its Hopf/ $B - L$ -exact argument is superseded — the Majorana mass is the $\Delta L = 2$ winding-changing process). Data layer (`bst_predictions.json`) and K740/K741 are authoritative.

Abstract

Bubble Spacetime Theory makes a sharp, falsifiable prediction for neutrinoless double-beta decay ($0\nu\beta\beta$). BST neutrinos are vacuum modes on the Shilov boundary $\check{S} = S^4 \times S^1$, with masses $m_1 = 0$, $m_2 = (7/12)\alpha^2 m_e^2/m_p = 0.00865$ eV, and $m_3 = (10/3)\alpha^2 m_e^2/m_p = 0.04940$ eV. The neutrino is Majorana: the light right-handed mode is non-unitary (the γ_5 shadow intertwiner is not invertible) and the odd embedding dimension $g = 7$ locks the chirality, so no Dirac partner exists and the Weinberg operator generates a Majorana mass. Consequently lepton number is violated by two units — the Majorana mass is precisely the $\Delta L = 2$ topology-changing process — and $0\nu\beta\beta$ occurs, with effective mass $|m_{\beta\beta}| = 3.63$ meV (Majorana phases zero) or $|m_{\beta\beta}| \in [1.44, 3.63]$ meV (phases unknown), in the normal ordering with $m_1 = 0$. This is testable: a confirmed $0\nu\beta\beta$ detection in the 1–4 meV window supports BST; a confirmed null result below ~ 1 meV (with an established $m_1 = 0$ hierarchical spectrum) falsifies it; and a signal clearly above ~ 4 meV would break the banked PMNS forms or the $m_1 = 0$ spectrum. The 1–4 meV floor is accessible to LEGEND-1000 combined with nEXO.

1. BST Neutrino Masses and Mixing (Review)

1.1 The Boundary Seesaw

In BST, neutrino masses arise from the boundary seesaw on $D_{IV}^5 = \text{SO}_0(5, 2)/[\text{SO}(5) \times \text{SO}(2)]$:

$$m_{\nu_i} = f_i \times \alpha^2 \times \frac{m_e^2}{m_p}$$

where:

- $m_e^2/m_p = 278.3$ eV is the BST seesaw scale (boundary mass squared divided by bulk mass)
- $\alpha^2 = 1/137^2$ provides the electroweak suppression (two Hopf fiber vertices)
- f_i are geometric factors from D_{IV}^5

The three mass eigenstates are:

Eigenstate	Factor f_i	BST mass (eV)	Observed (eV)
ν_1	0	0 (exact)	< 0.009
ν_2	$(n_C + 2)/(4N_C) = 7/12$	0.00865	≈ 0.00868
ν_3	$2n_C/N_C = 10/3$	0.04940	≈ 0.0503

BST predicts normal ordering ($m_1 < m_2 < m_3$) with $m_1 = 0$ exactly. The sum $\Sigma m_\nu = 0.058$ eV. [Updated 2026-08-13: the current bound is DESI DR2 + CMB < 0.064 eV (95%, Λ CDM; arXiv:2503.14738), not the stale 0.072. BST's 0.058 is safe by ~ 0.006 eV under Λ CDM. ★ COUPLING (T2561): BST's certified no-crossing dark energy ($w > -1$, T2559/T2560) forbids the phantom relaxation, so BST is committed to the tight Λ CDM bound and CANNOT retreat to the dynamical-DE < 0.16 eV. The boundary-corrected Feldman-Cousins < 0.053 eV breaches the NO oscillation floor 0.059 — a tension shared by all minimal-NO theories, not BST-specific. Cosmology now prefers positive Σm_ν at 2.7σ (favorable). The distinctive BST content is $m_1=0$ *exactly* + the specific m_2/m_3 + the $0\nu\beta\beta$ window, not the sum.]

1.2 PMNS Mixing Angles

All three PMNS angles are ratios of $n_C = 5$ and $N_C = 3$:

$$\sin^2 \theta_{12} = \frac{N_C}{2n_C} = \frac{3}{10} = 0.300$$

$$\sin^2 \theta_{23} = \frac{n_C - 1}{n_C + 2} = \frac{4}{7} = 0.5714$$

$$\sin^2 \theta_{13} = \frac{1}{n_C(2n_C - 1)} = \frac{1}{45} = 0.02222$$

The derived cosines:

$$\cos^2 \theta_{12} = \frac{7}{10}, \quad \cos^2 \theta_{13} = \frac{44}{45}$$

1.3 CP Phases

The Dirac CP phase is $\delta_{CP} = \pi$ to leading order (CP conservation in the leptonic sector).

The Majorana phases are $\alpha_{21} = \alpha_{31} = 0$ (see Section 3 for the geometric argument).

2. Dirac vs. Majorana: The BST Determination

This is the central question for $0\nu\beta\beta$: is the neutrino its own antiparticle?

2.1 The Standard Seesaw vs. the Boundary Seesaw

In the standard Type-I seesaw, the light neutrino mass arises from $m_\nu \sim m_D^2/M_R$, where m_D is a Dirac mass and M_R is a heavy right-handed Majorana mass. The seesaw necessarily produces Majorana neutrinos because the heavy partner violates lepton number.

In BST, the seesaw has a different geometric origin:

- m_e (boundary excitation on $\check{S} = S^4 \times S^1$) plays the role of m_D
- m_p (bulk excitation in $\pi_6 \subset A^2(D_{IV}^5)$) plays the role of M_R

But this is a geometric seesaw, not a Majorana seesaw. The electron and proton are distinct particles with distinct topological quantum numbers. The seesaw formula $m_\nu \sim \alpha^2 m_e^2 / m_p$ expresses a ratio of energy scales — the boundary-to-bulk mass hierarchy modulated by the electroweak coupling — without requiring either the neutrino or its antiparticle to carry Majorana mass.

2.2 The Neutrino on the Shilov Boundary

The neutrino is the vacuum quantum of D_{IV}^5 — the minimum excitation above the flat-connection vacuum (see `BST_VacuumQuantum_NeutrinoLambda.md`). Its properties:

- No S^1 winding: The neutrino carries zero electric charge. It has no topological winding number on the S^1 fiber.
- No color: The neutrino is a color singlet — it does not participate in the Z_3 closure of $\mathbb{CP}^2$.
- Weak coupling only: The neutrino interacts solely through the Hopf fibration $S^3 \rightarrow S^2$ on the Shilov boundary.

2.3 The Chirality Argument for Majorana (supersedes the earlier Dirac argument)

Charged particles in BST have definite S^1 winding numbers $n \in \mathbb{Z}$, with antiparticles carrying $-n$; the topological distinctness of particle and antiparticle is guaranteed by $\pi_1(S^1) = \mathbb{Z}$. For the neutrino, $n = 0$ (no charge), so the S^1 winding does not distinguish ν from $\bar{\nu}$ — the neutrino sits in the one sector where the winding protection is absent.

The determining fact is the chirality lock at odd $g = 7$ (K729/F571). The light right-handed neutrino mode is non-unitary: the γ_5 shadow intertwiner that would build a Dirac partner is not invertible, and the central volume element $\omega = \gamma_1 \cdots \gamma_7$ (central because g is odd) locks the internal chirality to spacetime chirality. There is therefore no light ν_R to pair with, and the Weinberg dimension-5 operator generates a Majorana mass for the left-handed neutrino (F413/K673).

What of the earlier Hopf argument (that $h(S^3 \rightarrow S^2) = 1$ protects a $\mathbb{Z}_2$ orientation and forbids Majorana)? It is superseded, and its own logic supplies the reframe. The Hopf orientation is protected against *continuous* deformation — but the Majorana mass is not a continuous deformation; it is precisely the non-perturbative $\Delta L = 2$ event that collapses the $\mathbb{Z}_2$. A topological invariant conserved perturbatively can still change by a discrete topology-changing process, and that process *is* the Majorana mass. The Hopf datum forbids a *smooth* ν - $\bar{\nu}$ identification; it does not forbid the discrete one that lepton-number-violating physics realizes.

Conclusion: BST neutrinos are Majorana. ν and $\bar{\nu}$ are identified by the $\Delta L = 2$ Majorana mass, the discrete winding-changing process the odd- g chirality lock makes unavoidable.

2.4 $B - L$ is violated by two units (the Majorana mass)

Lepton number is a topological count (the $SO(5)$ winding, T1945) — conserved perturbatively. The Majorana neutrino mass is the process that changes it: in $0\nu\beta\beta$, $\Delta L = 2$ and hence $\Delta(B - L) = 2$. So $B - L$ is not exactly conserved (this reverses the earlier “B–L exact via the Hopf invariant” claim of `BST_Conservation_Laws.md` §2.3/8.5, which is being revised in parallel). The winding-count conservation of L is broken by exactly the discrete $\Delta L = 2$ topology-change, and that is why $0\nu\beta\beta$ occurs — rare (it is a single non-perturbative event) but nonzero.

3. The Effective Majorana Mass: BST Computation

BST predicts Majorana neutrinos (§2.3), so this section computes the effective mass $|m_{\beta\beta}|$ from the banked BST masses and mixing angles — the primary $0\nu\beta\beta$ prediction. (Historically this computation was filed as the “Majorana fallback” to the retired Dirac reading; post-2026-07-16 it is the primary result.)

3.1 Definition

The effective Majorana mass is:

$$|m_{\beta\beta}| = \left| \sum_{i=1}^3 U_{ei}^2 m_i \right| = \left| |U_{e1}|^2 m_1 + |U_{e2}|^2 m_2 e^{i\alpha_{21}} + |U_{e3}|^2 m_3 e^{i(\alpha_{31}-2\delta_{CP})} \right|$$

where α_{21} and α_{31} are the Majorana phases and δ_{CP} is the Dirac CP phase. (We use the PDG convention where $U_{e3} = \sin \theta_{13} e^{-i\delta}$, so $U_{e3}^2 = \sin^2 \theta_{13} e^{-2i\delta}$.)

3.2 PMNS Matrix Elements

The electron-row elements squared:

$$|U_{e1}|^2 = \cos^2 \theta_{12} \cos^2 \theta_{13} = \frac{7}{10} \times \frac{44}{45} = \frac{308}{450} = 0.6844$$

$$|U_{e2}|^2 = \sin^2 \theta_{12} \cos^2 \theta_{13} = \frac{3}{10} \times \frac{44}{45} = \frac{132}{450} = \frac{22}{75} = 0.2933$$

$$|U_{e3}|^2 = \sin^2 \theta_{13} = \frac{1}{45} = 0.02222$$

Unitarity check: $308/450 + 132/450 + 10/450 = 450/450 = 1$. ✓

3.3 BST Prediction with Majorana Phases Zero

With $m_1 = 0$, $\alpha_{21} = 0$, and $\alpha_{31} = 0$ (BST leading-order values):

$$|m_{\beta\beta}| = |U_{e2}|^2 m_2 + |U_{e3}|^2 m_3$$

This separates into a solar term and a reactor term:

$$\text{Solar: } \frac{22}{75} \times \frac{7}{12} \times \alpha^2 \frac{m_e^2}{m_p} = \frac{77}{450} \times \alpha^2 \frac{m_e^2}{m_p} = 0.002536 \text{ eV}$$

$$\text{Reactor: } \frac{1}{45} \times \frac{10}{3} \times \alpha^2 \frac{m_e^2}{m_p} = \frac{2}{27} \times \alpha^2 \frac{m_e^2}{m_p} = 0.001098 \text{ eV}$$

The solar term dominates by a factor of 2.3. The total:

$$|m_{\beta\beta}|_{\text{BST-Majorana}} = \frac{331}{1350} \times \alpha^2 \frac{m_e^2}{m_p} = 3.63 \text{ meV}$$

where 331 is prime and $1350 = 2 \times 3^3 \times 5^2$.

3.4 Dependence on Majorana Phases

If the Majorana phases are unknown (as they would be in a generic Majorana scenario), the effective mass ranges over:

$$|m_{\beta\beta}| = \left| \frac{22}{75} m_2 e^{i\alpha_{21}} + \frac{1}{45} m_3 e^{i\alpha_{31}} \right|$$

- Maximum ($\alpha_{21} = 0, \alpha_{31} = 0$ or $\alpha_{21} = \pi, \alpha_{31} = \pi$): $|m_{\beta\beta}| = 3.63$ meV
- Minimum ($\alpha_{21} = \pi, \alpha_{31} = 0$ or $\alpha_{21} = 0, \alpha_{31} = \pi$): $|m_{\beta\beta}| = 1.44$ meV

The minimum is nonzero because the solar and reactor terms have different magnitudes and cannot completely cancel:

$$|m_{\beta\beta}|_{\min} = \left| \frac{22}{75} m_2 - \frac{1}{45} m_3 \right| = \left| \frac{77}{450} - \frac{2}{27} \right| \times \alpha^2 \frac{m_e^2}{m_p} = \frac{179}{1350} \times \alpha^2 \frac{m_e^2}{m_p} = 1.44 \text{ meV}$$

Key point: For normal hierarchy with $m_1 = 0$, there is no cancellation to zero. The effective Majorana mass has an absolute floor of ~ 1.4 meV regardless of Majorana phases.

3.5 Comparison with the CKM CP Phase Note

The BST derivation of CKM parameters (BST_CKM_CPPhase_Derivation.md) states that the “two Majorana phases = 0.” This is consistent with the BST picture: the Majorana phases would be zero because the Hopf fiber orientation is a topological datum, not a continuous phase. In a Majorana scenario, the phases α_{21} and α_{31} measure the relative CP transformation between mass eigenstates. Since BST vacuum modes (neutrino mass eigenstates) are distinguished by their geometric weight on D_{IV}^5 rather than by a complex phase, these phases vanish.

With $\alpha_{21} = \alpha_{31} = 0$, the BST-Majorana value is uniquely $|m_{\beta\beta}| = 3.63$ meV.

4. The BST Prediction

4.1 Primary Prediction: Majorana Neutrinos — $0\nu\beta\beta$ occurs

$$|m_{\beta\beta}| \in [1.44, 3.63] \text{ meV} \quad (\alpha_{21} = \alpha_{31} = 0 \Rightarrow 3.63 \text{ meV})$$

$0\nu\beta\beta$ occurs, in the 1–4 meV window, normal ordering with $m_1 = 0$. This prediction follows from:

1. The odd-g chirality lock: the light ν_R is non-unitary, so there is no Dirac partner (§2.3, K729/F413/K673).
2. The Weinberg dimension-5 operator generates a Majorana mass — the $\Delta L = 2$ process — so $B-L$ is violated by 2 (§2.4).
3. The banked BST masses ($m_1 = 0, m_2, m_3$) and PMNS angles ($\sin^2 \theta_{12}, \sin^2 \theta_{13}$), with the effective mass computed in §3.

A confirmed detection in $[1.44, 3.63]$ meV supports BST.

4.2 Retired reading: Dirac Neutrinos (superseded 2026-07-16)

The original BST prediction was $|m_{\beta\beta}| = 0$ (Dirac), from the Hopf-orientation topological argument and exact $B-L$ conservation (§2.3, §2.4 as originally written). This is superseded: the odd-g chirality lock forbids the light ν_R the Dirac reading required, and the Majorana mass is precisely the discrete $\Delta L = 2$ process the Hopf argument does not exclude. Recorded here for the audit trail; it is no longer a BST prediction. (*Note: a confirmed null result below ~ 1 meV — with an established $m_1 = 0$ hierarchical spectrum — would falsify the current Majorana prediction, not vindicate the retired Dirac one.*)

5. Comparison with Current and Future Experiments

5.1 Current Experimental Limits

Experiment	Isotope	Current Limit on $ m_{\beta\beta} $	Status
KamLAND-Zen 800	^{136}Xe	$< 36\text{--}156$ meV (90% CL)	Running
GERDA	^{76}Ge	$< 79\text{--}180$ meV	Completed
CUORE	^{130}Te	$< 90\text{--}305$ meV	Running
EXO-200	^{136}Xe	$< 147\text{--}398$ meV	Completed

The range in each limit reflects the nuclear matrix element (NME) uncertainty, which is the dominant systematic. All current limits are far above the BST-Majorana prediction of 3.63 meV.

5.2 Near-Future Experiments (2025–2030)

Experiment	Isotope	Target Sensitivity	Timeline
LEGEND-200	^{76}Ge	$\sim 20\text{--}50$ meV	Operating
KamLAND-Zen 800 (upgrade)	^{136}Xe	$\sim 20\text{--}50$ meV	Operating
CUPID	^{100}Mo	$\sim 10\text{--}20$ meV	~ 2028

These experiments will probe the inverted hierarchy band ($|m_{\beta\beta}| \sim 15\text{--}50$ meV). A null result at this scale is consistent with BST (which predicts normal ordering). A detection would falsify BST outright.

5.3 Next-Generation Experiments (2030+)

Experiment	Isotope	Target Sensitivity	Timeline
LEGEND-1000	^{76}Ge	$\sim 9\text{--}21$ meV	~ 2030
nEXO	^{136}Xe	$\sim 5\text{--}17$ meV	~ 2032
CUPID-1T	^{100}Mo	$\sim 5\text{--}15$ meV	> 2033

These experiments approach but may not reach the BST-Majorana floor of 3.63 meV. The combined reach of LEGEND-1000 + nEXO may push sensitivity to $\sim 3\text{--}5$ meV (depending on NME calculations), which would begin to probe the BST-Majorana prediction.

5.4 The Decisive Experiment

To definitively test the BST-Majorana fallback at $|m_{\beta\beta}| = 3.63$ meV requires an experiment with sensitivity below ~ 3 meV. This is beyond the projected reach of any currently funded experiment but is within the scope of proposed multi-ton-scale detectors. Such an experiment would resolve the question completely:

- Detection at ~ 3.6 meV: BST Dirac prediction falsified. BST-Majorana prediction confirmed (if Majorana phases near zero). BST masses and mixing validated.
- Detection at ~ 1.4 meV: BST Dirac prediction falsified. BST-Majorana with $\alpha_{21} \approx \pi$ confirmed.
- Null result below 1 meV (with $m_1 = 0$ hierarchical spectrum established): falsifies the BST Majorana prediction — the effective mass cannot fall below ~ 1.4 meV given the banked PMNS angles and $m_1 = 0$.

6. The BST Prediction in Context

6.1 Normal Hierarchy Floor

In any theory with normal hierarchy and $m_1 = 0$, the effective Majorana mass is bounded below by:

$$|m_{\beta\beta}|_{\min} = \left| |U_{e2}|^2 \sqrt{\Delta m_{21}^2} - |U_{e3}|^2 \sqrt{\Delta m_{31}^2} \right|$$

Using PDG 2024 best-fit values: $|m_{\beta\beta}|_{\min} \approx 1.2\text{--}1.5$ meV, $|m_{\beta\beta}|_{\max} \approx 3.5\text{--}4.2$ meV.

The BST values (1.44–3.63 meV) sit squarely within this standard-analysis band, as expected — BST’s masses and mixing angles are consistent with observation.

6.2 Inverted Hierarchy Exclusion

BST excludes inverted hierarchy (IH), which would give $|m_{\beta\beta}| \approx 15\text{--}50$ meV. A detection in the IH band would simultaneously:

- Establish Majorana neutrinos (falsifying BST’s Dirac prediction)
- Establish inverted ordering (falsifying BST’s normal ordering prediction)
- Produce a double falsification of BST

Conversely, a null result in the IH band (< 15 meV) would: (a) disfavor IH (consistent with BST), while (b) leaving the Dirac/Majorana question open pending sensitivity below 4 meV.

6.3 The Test

The cleanest formulation of BST’s $0\nu\beta\beta$ prediction:

$$\text{BST predicts: } 0\nu\beta\beta \text{ occurs, } |m_{\beta\beta}| \in [1.44, 3.63] \text{ meV (NO, } m_1 = 0\text{)}.$$

Every $0\nu\beta\beta$ experiment is a sharp test of BST:

- Confirmed detection in $[1.44, 3.63]$ meV (normal ordering, $m_1 = 0$): supports BST.
- Confirmed null below ~ 1 meV with an established $m_1 = 0$ hierarchical spectrum: falsifies BST.
- A signal clearly above ~ 4 meV: breaks the banked PMNS forms or the $m_1 = 0$ spectrum (a different falsification).

This makes $0\nu\beta\beta$ the single cleanest near-term test of BST, alongside the magnetic monopole search (MoEDAL, a Five-Absence bet) and the dark energy equation of state (DESI). Unlike the pre-2026-07-16 Dirac reading, a detection now supports BST rather than killing it.

7. Half-Life Estimates

For reference, the $0\nu\beta\beta$ half-life is related to $|m_{\beta\beta}|$ by:

$$[T_{1/2}^{0\nu}]^{-1} = G_{0\nu}(Q, Z) \times |M_{0\nu}|^2 \times \left(\frac{|m_{\beta\beta}|}{m_e} \right)^2$$

where $G_{0\nu}$ is the phase space factor and $|M_{0\nu}|$ is the nuclear matrix element. At the BST-Majorana value $|m_{\beta\beta}| = 3.63$ meV:

Isotope	$G_{0\nu}$ (yr ⁻¹)	NME range	$T_{1/2}$ (yr)
⁷⁶ Ge	2.36×10^{-15}	2.8–6.1	$\sim 10^{29}–10^{30}$
¹³⁶ Xe	1.45×10^{-14}	1.6–3.7	$\sim 10^{29}–10^{30}$

These half-lives are 1–2 orders of magnitude beyond the reach of LEGEND-1000 and nEXO, confirming that detection at the BST-Majorana level will require a subsequent generation of experiments.

8. Connection to Other BST Results

8.1 The Vacuum Quantum

The neutrino is the vacuum quantum of D_{IV}^5 (BST_VacuumQuantum_NeutrinoLambda.md). The lightest neutrino ν_1 IS the vacuum ($m_1 = 0$ exactly). The massive modes ν_2 and ν_3 are the vacuum fluctuating between geometric modes.

For $0\nu\beta\beta$, the relevant question is whether the vacuum can be its own anti-vacuum. In BST, the vacuum is unique — there is one vacuum state, not a particle-antiparticle pair. This uniqueness is another route to the Dirac conclusion: the vacuum has no antiparticle, so ν_1 (which IS the vacuum) cannot be Majorana. The massive modes inherit this structure.

8.2 The 1920 Cancellation and the Proton Mass

The proton mass $m_p = 6\pi^5 m_e$ enters the seesaw denominator. The factor $6\pi^5$ arises from the Casimir eigenvalue $C_2 = 6$ and the Bergman volume $\pi^5/1920$, with the 1920 cancelling between the Hua volume formula and the baryon circuit orbit (BST_BaryonCircuit_ContactIntegral.md). The neutrino mass is therefore:

$$m_\nu = f \times \frac{\alpha^2}{6\pi^5} \times m_e = f \times \alpha^{14} \times m_{p1}$$

The exponent $14 = 2 \times \text{genus}$ connects the neutrino mass to the cosmological constant via $\Lambda \propto \alpha^{56} = (\alpha^{14})^4 \propto m_\nu^4$.

8.3 The Neutrino-Lambda Connection and $0\nu\beta\beta$

If $0\nu\beta\beta$ were observed, confirming Majorana neutrinos, the BST relationship $\Lambda \propto m_\nu^4$ would remain valid (it depends on the mass values, not on Dirac/Majorana nature). However, the topological structure of BST — specifically the Hopf fiber argument for $B - L$ conservation — would be falsified, requiring a fundamental revision of how BST treats the Shilov boundary.

9. What Is Proved vs. Open

Established

Claim	Status	Reference
$m_1 = 0, m_2 = 0.00865 \text{ eV}, m_3 = 0.04940 \text{ eV}$	Derived (0.35–1.8%)	BST_NeutrinoMasses.md
Normal ordering	Predicted	BST_NeutrinoMasses.md
PMNS angles from n_C, N_C	Derived (0.1–2.3%)	BST_CKM_PMNS_MixingMatrices.md
$\delta_{CP} = \pi$ (Dirac phase)	Predicted	BST_CKM_PMNS_MixingMatrices.md
Majorana phases = 0	Predicted	BST_CKM_CPPhase_Derivation.md

Claim	Status	Reference
$B - L$ violated by $\Delta L = 2$ (Majorana mass)	Predicted (rev. 2026-07-16)	BST_Conservation_Laws.md (revising)
Neutrinos are Majorana	Predicted (F413/K673)	This note; K740/K741
$\ m_{\beta\beta}\ \in [1.44, 3.63]$ meV	Predicted	This note
$0\nu\beta\beta$ occurs	Predicted	This note

Open

Question	Status	Priority
Fix the Majorana phases α_{21}, α_{31} from the geometry (narrow $[1.44, 3.63]$ to a point)	Open	1
Geometric origin of $f_3 = 10/3$ and $f_2 = 7/12$	Conjectured	2
Is the $\Delta L = 2$ scale (the Weinberg operator coefficient) fixed by the geometry?	Open	3
NME calculations for BST-specific nuclear structure	Standard physics	4

10. Summary

BST makes a clean, sharp prediction (revised 2026-07-16, Dirac $\rightarrow$ Majorana):

$$0\nu\beta\beta \text{ occurs. Neutrinos are Majorana. } |m_{\beta\beta}| \in [1.44, 3.63] \text{ meV.}$$

This follows from the odd-g chirality lock: the light ν_R is non-unitary, so no Dirac partner exists, and the Weinberg operator generates a Majorana mass — the $\Delta L = 2$ topology-changing process that violates $B - L$ by two units. The masses ($m_1 = 0, m_2, m_3$) and PMNS angles are the banked BST forms.

The value $|m_{\beta\beta}| \in [1.44, 3.63]$ meV sits at the bottom of the normal-hierarchy band, below the inverted-hierarchy floor (~ 15 meV), and below current single-experiment sensitivity. Combined next-generation reach (LEGEND-1000 + nEXO, $\sim 1\text{--}4$ meV) is required.

The experimental program:

1. 2025–2028 (LEGEND-200, KamLAND-Zen): probe the IH band; a null there is consistent with BST’s NO, $m_1 = 0$ spectrum.
2. 2028–2033 (LEGEND-1000, nEXO, CUPID): probe to $\sim 5\text{--}10$ meV.
3. 2033+ (multi-ton / combined): probe the $1\text{--}4$ meV floor. The decisive test: a detection in $[1.44, 3.63]$ meV supports BST; a confirmed null below ~ 1 meV (with $m_1 = 0$ established) falsifies it.

A confirmed detection in the $1\text{--}4$ meV window supports BST — the reverse of the retired Dirac reading, which is why propagating this revision through the corpus matters (K740).

Research note, March 2026 — revised 2026-07-18 (Dirac $\rightarrow$ Majorana, per the 2026-07-16 reversal F413/K673; see K740/K741). Casey Koons & the CI team (Lyra, Keeper, Elie, Grace). For the BST GitHub repository.